tt-analysis

Check point commit -> BEFORE saving the proton related variables in the real data samples

Plots

- Made with plot_m_new.cpp
- samples used are:
  - DY_2018_UL_MuTau_nano_merged_pileup_protons.root
  - ttjets_2018_UL_MuTau_nano_merged_pileup_protons.root
  - QCD_2018_UL_MuTau_nano_merged_pileup_protons.root
  - Data_2018_UL_MuTau_nano_merged_pileup_protons.root
  - MuTau_sinal_SM_2018_july.root -> signal file
- The processing for these files is the same as done in original analysis;
- For the samples where the proton related variables are saved from the beginning the number of events **should be the same**
  - this was done because adding pileup protons to real data was not the most correct procedure

The Data and QCD samples in these plots have added pileup protons to them

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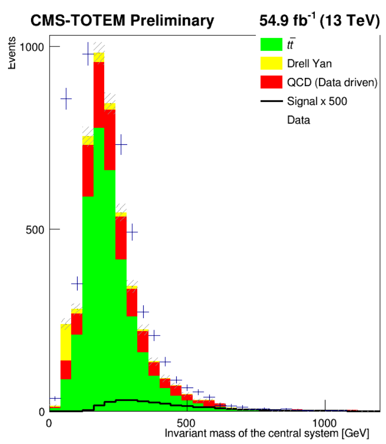

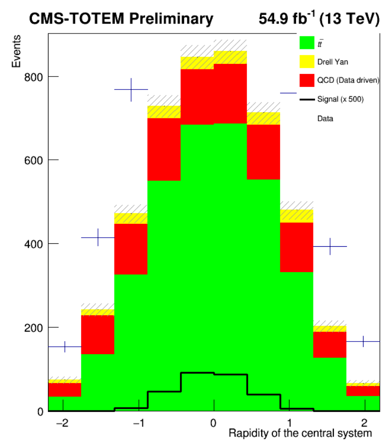

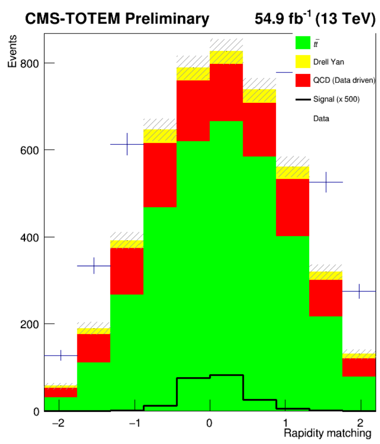

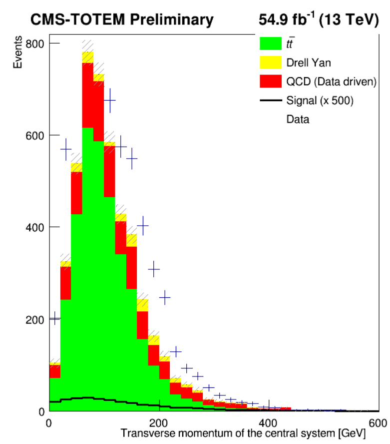

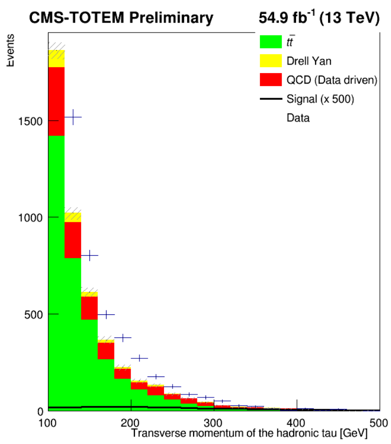

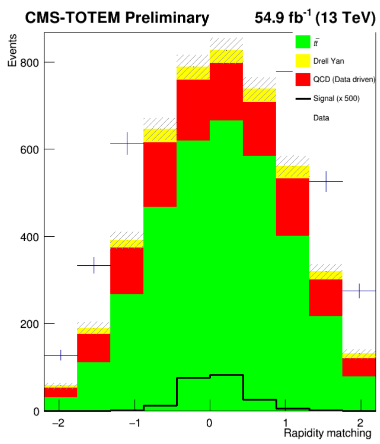

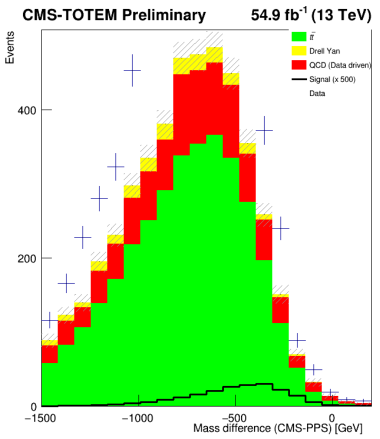

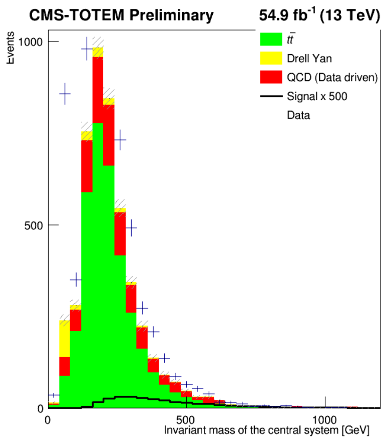

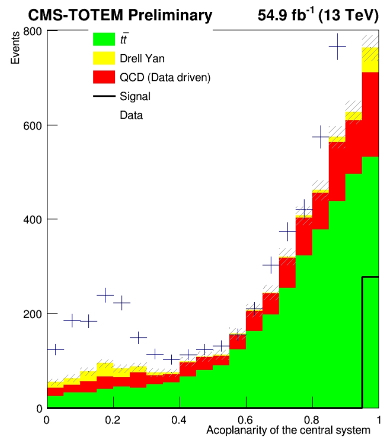

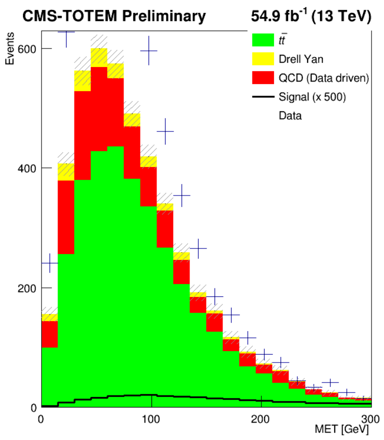

Cuts applied

MuTau Channel

Category / Cut	ttjets	dy	qcd	data
FASE0	1 tau, 1 muon	1 tau, 1 muon	1 tau, 1 muon	1 tau, 1 muon
Double_medium				
Iso_Mu24	✓	✓	✓	✓
Ele32_WpTight				
Electron_				
lumi filtering			✓	✓
FASE1				
tau id1	>63	>63	>63	>63
tau id2	>7	>7	>7	>7
tau id3	>1	>1	>1	>1
tau pt	>100	>100	>100	>100

Category / Cut	ttjets	dy	qcd	data
mu id	≥3	≥3	≥3	≥3
mu pt	>35	>35	>35	>35
ele id				
ele pt				
delta R	>0.4	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4	<2.4
charge signs	opposite	opposite	same	opposite
scaling factors	✓	✓		
valid jet and MET info	✓	✓		

Nr of events

	Starting sample	Phase0	Phase1
Data **	984,127,247	698,931,141	1,708
QCD **	984,127,247	698,931,141	234
Data	984,127,247	19,698	6,859
QCD	984,127,247	19,698	1,011
ttjets	304,895,029	42,500,241	26,649
DY	195,510,810	41,388,654	1,704
signal	299,979	NA	13,893

** Is the samples that save the proton related variables instead of adding pileup protons later.

--> Looks like the issue is the change in code to save proton related variables in phase1

TODO: re-run analysis ; back to original code and then go from there